

Part F: End-to-End Validation

Full workflow run on a local **kind** cluster, following the exact sequence from the assignment.

1. Apply manifests + train

```
$ kubectl apply -f k8s/namespace.yaml
namespace/ml-training created

$ kubectl apply -f k8s/configmap.yaml
configmap/training-config created

$ kubectl apply -f k8s/pvc.yaml
persistentvolumeclaim/training-data-pvc created
persistentvolumeclaim/checkpoints-pvc created

$ kubectl apply -f k8s/training-job.yaml
job.batch/model-training created
```

2. Training completes (real, full CIFAR-10 dataset)

```
$ kubectl logs job/model-training -n ml-training
Files already downloaded and verified
Files already downloaded and verified
{"epoch": 1, "train_loss": 1.3687, "train_accuracy": 0.499, "val_loss": 1.158, "val_accuracy": 0.6032}
{"event": "checkpoint_saved", "path": "/app/checkpoints/classifier_v1.pt"}
{"event": "training_complete", "best_val_loss": 1.158}

$ kubectl get jobs -n ml-training
NAME                STATUS      COMPLETIONS   DURATION
model-training      Complete    1/1           5h50m
```

3. Deploy serving layer

```
$ kubectl apply -f k8s/serving-deployment.yaml
deployment.apps/model-serving created

$ kubectl apply -f k8s/serving-service.yaml
service/model-serving created

$ kubectl apply -f k8s/hpa.yaml
horizontalpodautoscaler.autoscaling/model-serving created
```

4. Verify pods are running and healthy

```
$ kubectl get pods -n ml-training
NAME                                READY   STATUS    RESTARTS   AGE
model-serving-64d79654d7-25rmv     1/1     Running   0          13m
model-serving-64d79654d7-gkcxv     1/1     Running   0          13m
model-training-r982t                0/1     Completed 0          16m

$ kubectl describe deployment model-serving -n ml-training
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
```

5. Test the prediction endpoint

```
$ kubectl port-forward svc/model-serving 8080:80 -n ml-training &
Forwarding from 127.0.0.1:8080 -> 8080

$ curl http://localhost:8080/health
{"status":"ok"}
$ curl -X POST http://localhost:8080/predict -F "image=@test_image.png"
```

```
{"predictions":[
  {"class":"ship","probability":0.195266},
  {"class":"truck","probability":0.16377},
  {"class":"automobile","probability":0.113274},
  {"class":"cat","probability":0.103262},
  {"class":"horse","probability":0.102552},
  {"class":"airplane","probability":0.085738},
  {"class":"bird","probability":0.071926},
  {"class":"deer","probability":0.06473},
  {"class":"frog","probability":0.051797},
  {"class":"dog","probability":0.047685}
]}
```

Issues found and fixed during validation

1. **NumPy/PyTorch ABI mismatch** - torch==2.2.2 requires numpy<2. Without pinning it, pip installed NumPy 2.x and every operation touching image tensors crashed with *RuntimeError: Numpy is not available*. Fixed by pinning numpy==1.26.4 in requirements/train.txt and requirements/serve.txt.
2. **CPU thread oversubscription** - PyTorch defaulted its thread pool to the node's full CPU count (12) instead of the container's 2-CPU cgroup limit, causing severe scheduling contention. Fixed by setting OMP_NUM_THREADS/MKL_NUM_THREADS=2 in k8s/training-job.yaml.
3. **Hardware constraint** - CPU-only training on a 2-core limit took roughly 45 minutes to an hour per epoch on the full CIFAR-10 dataset.

Screenshots (real terminal output)

Job complete, pods running, deployment describe:

```
Terminal Shell Edit View Window Help
arpit.singh -- -zsh - kubecti -- 244x63
...age8-api-all--jogback/batchman -- -zsh
~ -- -zsh - kubecti
~ -- -zsh
...
~ -- arpit-singh@stage8-1c-api -- -zsh
...e7-api-all--jogback/balance-api -- -zsh
+

curl http://localhost:8888/health
curl -X POST http://localhost:8888/predict -F "image=@tmp/test_image.png"

Switched to context "kind-mlops-pytorch-pipeline".
NAME          STATUS    COMPLECTIONS  DURATION  AGE
model-training Complete  1/1          5h50m    20h
Files already downloaded and verified
Files already downloaded and verified
{"epoch": 1, "train_loss": 1.3687, "train_accuracy": 0.499, "val_loss": 1.158, "val_accuracy": 0.6832}
{"event": "checkpoint_saved", "path": "/app/checkpoints/classifier_v1.pt"}
{"event": "training_complete", "best_val_loss": 1.158}
NAME          STATUS    RESTARTS  AGE
model-serving-64d79654d7-26rsv 1/1      Running   0      20h
model-serving-64d79654d7-gkxcb 1/1      Running   0      20h
model-training-9mgg6           0/1      Completed 0      20h
Name:          model-serving
Namespace:     ml-training
CreationTimestamp: Sat, 29 Aug 2026 16:25:53 +0530
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 1
Selector:      app=model-serving
Replicas:      2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 0 max unavailable, 1 max surge
Pod Template:
Labels: app=model-serving
Containers:
  serving:
    Image:      mlops-serve:v1
    Port:       8888/TCP
    Host Port:  0/TCP
    Limits:
      cpu:      1
      memory: 2Gi
    Requests:
      cpu:      500m
      memory: 1Gi
    Liveness:    http-get http://:8888/health delay=5s timeout=10s #success=1 #failure=3
    Readiness:   http-get http://:8888/health delay=15s timeout=10s #success=1 #failure=3
    Environment: <none>
    Mounts:
      /app/checkpoints from checkpoints-volume (ro)
      /app/configs from config-volume (ro)
Volumes:
  checkpoints-volume:
    Type:      PersistentVolumeClaim (a reference to a PersistentVolumeClaim in the same namespace)
    ClaimName: checkpoints-pvc
    ReadOnly:  true
  config-volume:
    Type:      ConfigMap (a volume populated by a ConfigMap)
    Name:      training-config
    Optional:  false
Node-Selectors: <none>
Tolerations:   <none>
Conditions:
  Type              Status  Reason
  ----              -
  Available          True    MinimumReplicasAvailable
  Progressing        True    NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet:  model-serving-64d79654d7 (2/2 replicas created)
```

Health check and predict response:

```
Terminal Shell Edit View Window Help
arpit.singh -- zsh - kubect! - 244x63
...age8-api-all--logback/batchman -- zsh ... -- zsh - kubect! ... -- zsh ... -- arpit-singh@stage8-tc-api -- zsh ... ..e7-api-all--logback/balance-api -- zsh ... +

{"event": "training_complete", "best_val_loss": 1.159}
NAME          READY   STATUS    RESTARTS   AGE
model-serving-64d79654d7-25rmv  1/1     Running   0           28h
model-serving-64d79654d7-gkxzb  1/1     Running   0           28h
model-training-9mgg6            0/1     Completed 0           28h
Name:          model-serving
Namespace:     ml-training
CreationTimestamp: Sat, 29 Aug 2026 16:25:53 +0530
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 1
Selector:      app=model-serving
Replicas:      2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 0 max unavailable, 1 max surge
Pod Template:
  Labels: app=model-serving
  Containers:
    serving:
      Image:      mlops-serve:v1
      Port:       8888/TCP
      Host Port:  0/TCP
      Limits:
        cpu:      1
        memory:  2Gi
      Requests:
        cpu:      500m
        memory:  1Gi
      Liveness:    http-get http://:8888/health delay=0s timeout=1s period=10s #success=1 #failure=3
      Readiness:  http-get http://:8888/health delay=15s timeout=1s period=5s #success=1 #failure=3
      Environment: <none>
      Mounts:
        /app/checkpoints from checkpoints-volume (ro)
        /app/configs from config-volume (ro)
  Volumes:
    checkpoints-volume:
      Type:      PersistentVolumeClaim (a reference to a PersistentVolumeClaim in the same namespace)
      ClaimName: checkpoints-pvc
      ReadOnly:  true
    config-volume:
      Type:      ConfigMap (a volume populated by a ConfigMap)
      Name:      training-config
      Optional:  false
    Node-Selectors: <none>
    Tolerations:   <none>
  Conditions:
    Type      Status  Reason
    ----      -
    Available  True    MinimumReplicasAvailable
    Progressing True    NewReplicaSetAvailable
    OldReplicaSets: <none>
    NewReplicaSet:  model-serving-64d79654d7 (2/2 replicas created)
  Events:
    [2] 16085
  Unable to listen on port 8888: Listeners failed to create with the following errors: [unable to create listener: Error listen tcp4 127.0.0.1:8888: bind: address already in use unable to create listener: Error listen tcp6 :::1:8888: bind: address already in use]
  error: Unable to listen on any of the requested ports: [8888 8889]
  [2] + exit 1   kubect! port-forward svc/model-serving 8888:88 -n ml-training
  Handling connection for 8888
  {"status": "ok", "handling connection for 8888
  {"predictions": [{"class": "ship", "probability": 0.195266}, {"class": "truck", "probability": 0.16377}, {"class": "automobile", "probability": 0.113274}, {"class": "cat", "probability": 0.183262}, {"class": "horse", "probability": 0.192552}, {"class": "airplane", "probability": 0.085738}, {"class": "bird", "probability": 0.071926}, {"class": "deer", "probability": 0.06473}, {"class": "frog", "probability": 0.051797}, {"class": "dog", "probability": 0.047685}]}
  arpit.singh@TC-10044 ~ %
```